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RESEARCH ARTICLE

CLEP-Based EEG to Text Reconstruction

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ABSTRACT With the growing demand for Brain-Computer Interface (BCI) systems among patients with paralysis and aphasia, the challenge of recovering semantic information from language-related EEG signals urgently needs to be addressed, and research on tonal languages remains in its early stages. To address this, we propose a continuous Chinese semantic reconstruction model utilizing EEG signals to enhance system performance, and propose a BCI pre training framework based on the comparison of language and EEG signals. In this framework, CLEP is designed to enhance the accuracy of text reconstruction, while ChatGLM-6B ensures the continuity and coherence of the generated text. The framework was trained on a server using a publicly available Chinese EEG dataset, in addition to data collected from three subjects. The Experimental results were evaluated using two metrics, BERTScore and BLEU. The scores demonstrate that the framework outperforms random baseline tests. In particular, the accuracy of BERTScore score on the self collected dataset has been improved by 0.9%, and the effectiveness of the model has been demonstrated through Z-test. In addition, the feature capture ability of the CLEP model and the authenticity of human brain region responses were verified through feature extraction and visualization experiments.

INDEX TERMS Electroencephalographic, cross-modal training, brain-computer interface, semantic recovery, text reconstruction.

I. INTRODUCTION

Language is the most important tool for people to establish connections with the world. The mechanism by which the brain generates language has intrigued scientists for centuries [1]. This issue involves multiple disciplines, including brain science, neuroscience, cognitive science, and biomedical engineering. In recent years, brain-computer interface (BCI) technology has gained significant prominence, offering a novel technical approach to address this challenge. The current mainstream paradigm of brain-computer interface technology is divided into three categories: SSVEP, MI, and P300. These three categories use different forms of brainwave induction and, through decoding algorithms, have found applications in areas such as emotion classification [2], gaming [3], and robot control [4].

Meanwhile, the success of deep learning (DL) in image classification has motivated researchers to explore its

application in BCIs. From the current research results, there are already many examples showing that combining DL with BCI improves the performance of the model, and has brought great help to patients with paralysis. This progress has prompted researchers to address a wider spectrum of needs, such as restoring motor functions in paralyzed patients and assisting individuals with communication impairments. BCIs offer a novel communication pathway for these populations.

In the field of semantic restoration based on brain signals, a considerable number of researchers have been involved in this work.

Currently, more researchers have adopted invasive BCI techniques to carry out semantic reconstruction studies of the human brain, and the signals they use include ECoG, sEEG, etc. Due to the invasive risks associated with surgery and the difficulties in selecting subjects, most people prefer to accept non-invasive acquisition of brain signals, which include fMRI, MEG, and EEG. Among them, EEG is a relatively easy to acquire signal with high temporal resolution.

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On the other hand, most of the international research in this field has only focused on the recovery of non-tonal languages from the human brain, whereas about 60% of the world's languages are tonal languages, and Chinese is one of them, and work in this area needs to be promoted.

The main work of this study is based on the recovery of Chinese text from the human brain using EEG signals. The main contributions of this research are as follows:

1. A study was conducted to construct an EEG dataset under Chinese audio stimulation.
2. A BCI framework for Chinese text recovery based on EEG was developed.
3. The CLIP model has been improved to enable cross-modal training of EEG and text signals.
4. Based on this framework, there is a significant improvement in the accuracy of text recovery on self-collected dataset.

II. RELATED WORK

Although existing augmentative and alternative communication (AAC) technologies, such as head or eye trackers, are available, they suffer from low information transfer rates and become increasingly difficult to operate as patients lose voluntary muscle control [5].

A. INVASIVE BCI

Currently, research has shown that different brain regions exhibit different reactions when reading aloud, listening to audio, or even imagining listening and attempting to pronounce [6], [7], [8]. Based on this, researchers have utilized invasive BCI techniques to obtain high-quality brain signals, helping aphasia patients regain their communication skills.

Based on the paradigm of vocalization: As early as 2015, Christian Herff et al. made preliminary explorations in decoding continuous spoken speech from neural matrices related to speech and language processing [9]. In recent years, Gopala K. Anumachipalli et al. have designed a neural decoder that explicitly utilizes encoded kinematics and sound representations to synthesize audible speech [10].

Under the same paradigm, Joseph G. Makin et al. modeled the ECoG to text reconstruction problem as a machine translation problem, treating brain signals as source language and corresponding continuous text as target language, thus transferring the machine translation model to the task of decoding brain activity [11].

Based on the paradigm of attempting to speak out: Unlike the above paradigm, David A. Moses et al. used deep learning algorithms to create computational models for detecting and classifying words from recorded cortical activity patterns. As a result, it decoded sentences in real-time at an average speed of 15.2 words per minute, with a median word error rate of 25.6% [12].

Under the same paradigm, Nicholas S. Card et al. implanted four microelectrode arrays into the left central anterior gyrus of a patient with amyotrophic lateral sclerosis (ALS) and recorded the neural activity of 256 cortical

electrodes. The article reported the results of decoding cortical neural activity when he attempted to speak in a prompt and unstructured conversational environment. 25 days after surgery, with an increase in recorded EEG training data, the neural prosthesis maintained an accuracy rate of 97.5% using 125000 words, with a cumulative time of over 248 hours [13].

The paradigm of imagining handwriting movements: Under the paradigm of imagining handwritten movements, Francis R. Willett et al. developed an invasive BCI that decodes imagined handwritten movements from neural activity in the motor cortex. Using this BCI system, research participants paralyzed due to spinal cord injury achieved a typing speed of 90 characters per minute, with an online raw accuracy rate of 94.1% [14].

Despite the fact that high-quality ECoG signals can be obtained using invasive BCI, the disadvantages are obvious. The selection of subjects is more difficult with this technique, and there is a risk that the electrodes will deteriorate and lead to infected wounds after the electrode implantation procedure.

B. NON-INVASIVE BCI

As a result, an increasing number of researchers have conducted research on non-invasive BCI-based semantic reconstruction of the human brain.

1) MEG

With the wearable nature of MEG acquisition devices [15], Debadatta Dash et al. studied the decoding of five imaginative and spoken phrases from a single trial, non-invasive magnetoencephalography (MEG) signals collected from eight adult subjects. The experimental results indicate that imaginative and spoken phrases can be directly decoded from neural magnetic signals. CNN is very effective, with an average decoding accuracy of 93% for imaginative phrases and 96% for spoken phrases [16].

2) fMRI

Another non-invasive brain signal recording method is fMRI. This paper is also the baseline of our work. Jerry Tang et al. attempted to recover the auditory stimuli that the human brain is receiving or imagining through fMRI signals, and they used MFA to align the audio and text and then aligned the fMRI signals from there. In the encoding model stage, the article used the ridge regression model to complete the fitting between the data, and to ensure the accuracy of semantic recovery in the inference stage, the article introduces a large language model for text prediction and adopts a beam search generation strategy to provide candidate words, which is combined with the trained fitting model to select the optimal candidate words to retain the most probable continuation of the text, thus realizing the semantic reconstruction of the human brain. The results show that the accuracy of the generated text is significantly higher than expected by chance, which proves the effectiveness of the method. However, in the

fitting stage, the article uses the traditional machine learning method of ridge regression, which makes it difficult to fit the two types of data better, which limits the performance of the model [17]. The model inspires the research work in this paper.

3) EEG

Compared to fMRI, EEG is more convenient to obtain. Zhenhailong Wang et al. regarded the human brain as a special encoder and used the pre-trained language model BART for continuous text decoding of EEG signals for the first time. Based on this, the article proposes a new framework called BrainBART. This work utilized the ZuCo dataset, and BrainBART achieved a BLEU-1 score of 40.1% in continuous text decoding [18]. The research did not make any internal modifications to the BART model; it only carried out front-end work to align the modalities.

Yiqun Duan et al. introduced a new framework called DeWave. It integrates discrete encoding sequences into the open vocabulary EEG to text translation task. This work is the first to promote the translation of the entire EEG signal cycle without word-level sequential labeling (e.g. eye gaze), achieving 20.5 BLEU-1 and 29.5 Rouge-1 on the ZuCo dataset [19].

Meanwhile, Nuwa Xi et al. proposed a generalized framework for decoding brain signals, UniCoRN. UniCoRN achieved a BLEU-4 score of 37.04% on the continuous text decoding task using EEG from the ZuCo dataset, and achieved a BLEU-4 score of 62.90% on the continuous text decoding task using EEG from the ZuCo dataset. [20]. The above models provide us with certain references in terms of structure, but none of them have made attempts in the restoration of tonal language.

C. RESEARCH ON RESTORATION BASED ON NON-TONAL LANGUAGES

The above studies were all conducted on restoring text or nontonal languages. However, over 60% of languages worldwide are tonal, and due to the differences in pronunciation and brain responses between tonal and nontonal languages [21], [22], Based on this, recovery models for non tonal languages should be independently studied. Yan Liu and Daohan Zhang have made attempts to decode pronunciations and long sentences from neural activity in tonal language [23], [24]. The article distinguishes characters and tones separately, which increases the difficulty of the task.

For the reconstruction of a tonal language, Hui Zheng et al. collected a well-annotated Chinese word reading sEEG dataset from 12 subjects, targeting language-related brain networks. Using this benchmark test, they developed the Du-IN1 model, which extracts context embeddings based on region-level labeling through discrete codex-guided mask modeling. This model achieved state-of-the-art performance on a classification task of 61 words, surpassing all baselines [25]. This model is unable to restore long texts.

While the above research works to restrict the candidate words to a smaller thesaurus, this paper attempts to recover Chinese text from EEG signals by combining the advantages of ChatGLM-6B, a large language model, on an open thesaurus. In addition, in this literature, the author reveals that language functional areas are not single and isolated brain regions. Language processing is a distributed, whole brain collaborative process, which supports our approach of using whole brain EEG data to extract corresponding brain region weights during training [26].

III. METHOD

A. DATA

There are two sources of data used in this paper: the public dataset Chinese EEG and the self-collected dataset. Among them, WLB, JQD, and LW represent the code names of three subjects. The commonality between the two datasets is that both utilize Chinese language materials in their respective paradigms. The difference lies in the length of each piece of material, while the public dataset is shorter. Moreover, the public dataset adopts a paradigm based on gaze text, while the self-acquisition experiment is based on a paradigm based on listening materials.

Public dataset Chinese EEG: In this paper, we used the Chinese EEG, a publicly available dataset jointly collected by the Southern University of Science and Technology and the University of Macau, which contains recordings of the EEG signals of 15 subjects during silent reading of textual materials. The researchers used eye-tracking techniques during data acquisition and EEG preprocessing methods such as filtering and ICA after acquisition to ensure strict alignment of text and EEG signals and purity of EEG signals [27]. In this paper, we selected the EEG data of three subjects as experimental data.

Self-collected dataset: On the other hand, this paper uses a self-collected dataset. The study selected three participants and developed an experimental paradigm while listening to audio materials, which required participants to sit on chairs at a certain distance from a screen with a fixed cross symbol. After the experiment begins, participants will have three minutes of preparation time, after which they will hear audio ranging from 4s to 16s, with a 3-second interval between each audio, as a brief rest time for the participants. The experiment recorded EEG data from 18 sessions in 59 channels. The channel configuration is shown in Figure 1. The color intensity in the figure represents the magnitude of the impedance values. Before the experiment began, we ensured that the impedance values were all below 50.

Each session containing three runs, each run containing 60 trials, and each run lasting 9-16 minutes. After the experiment, the experimenters will ask the subjects questions based on the audio content to confirm that the subjects were focused during the experiment. The experimental equipment used is the NeuSen W product provided by Neuracle, with a sampling rate of 1000Hz. The experimental paradigm flowchart of the

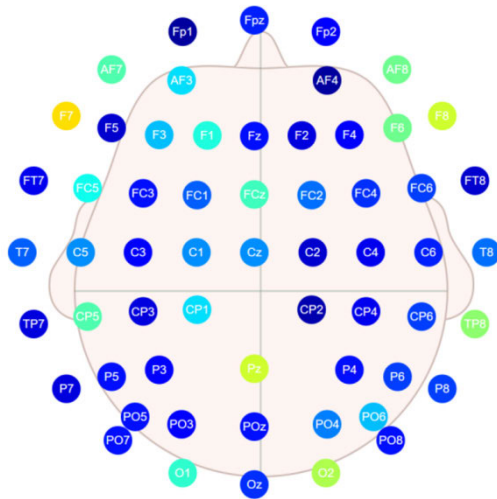


FIGURE 1. Channel configuration of self-collection experimental paradigm.

entire self-collected data is shown in Figure 2. The experimental protocol was approved by the relevant Institutional Review Board (IRB) or ethics committee.

Part of the text materials for the self-collected experiment (Session1-Session5) are from the publicly available dataset data_thchs30 at Tsinghua University, while the other part (Session6-Session18) is generated by AI. The experiment creates text into audio through code and conversion tools. During the experiment, the study used code to synchronize audio playback with brain signal recording, and then performed simple preprocessing such as filtering on EEG signals. The data structure of the partially collected EEG (Sessions 9-Sessions 11) is shown in Table 1.



FIGURE 2. Flow chart of self-collected data experimental paradigm.

B. DATA PREPROCESSING

In the preprocessed version of the public dataset Chinese EEG, EEG data with a sampling rate of 256Hz after bandpass filtering and downsampling at 0.5-30Hz were selected as experimental data; In the preprocessing of self-collected data, the experiment first divided the EEG data from each complete run into small segments of audio with a duration of 4-16s. Then, the experiment created a dictionary with text values and EEG data paths as keys as the data source for the training process. Before feeding the EEG data into the model, we mainly carried out three aspects of work: 1. We checked whether there were any bad sectors or serious time information deficiencies in the test sessions. If so, we removed the test session. 2. The data is filtered and downsampled in the

order of filtering first and then downsampling. This sequence can avoid the possible spectral aliasing interference with the original signal caused by downsampling. When filtering, the method we adopt is an fir finite impulse response filter, and the passband is set to 0.5-40Hz. 3. We perform downsampling at 1000Hz to 250Hz on the filtered data. The main purpose of downsampling is to reduce the subsequent computational complexity.

C. BCI FRAMEWORK BASED ON EEG SEMANTIC RECONSTRUCTION

The BCI framework based on EEG semantic reconstruction is shown in Figure 3. The system internally includes a CLEP module and a ChatGLM-6B module. The inputs to CLEP are: a candidate character set (the top five characters with the highest probability from the output of ChatGLM-6B, which must be manually set during the initial round) and the corresponding preprocessed EEG signal segments. The output of CLEP is: a single character (the candidate character with the highest probability as determined by CLEP). The input to the ChatGLM-6B module is: the accumulated outputs of CLEP (a single character in the first round, followed by two characters, three characters, and so on). The output of the ChatGLM-6B module is: a candidate pool of the next character that continues the input text (the top five candidate characters from the pool are selected as the text input for the next round of CLEP).

D. ENCODER

CLIP (Contrastive Language-Image Pre-Training) is a model that learns visual concepts from natural language supervision. [28]. In this paper, a modified version of the CLIP model, Contrastive Language-EEG Pre-Training (CLEP), is used to construct the EEG and text encoder, which uses the cross-modal learning of contrast between data from EEG to text to accomplish the mapping between EEG and text. CLEP is divided into three main parts: the text encoder, EEG encoder, and contrast training module.

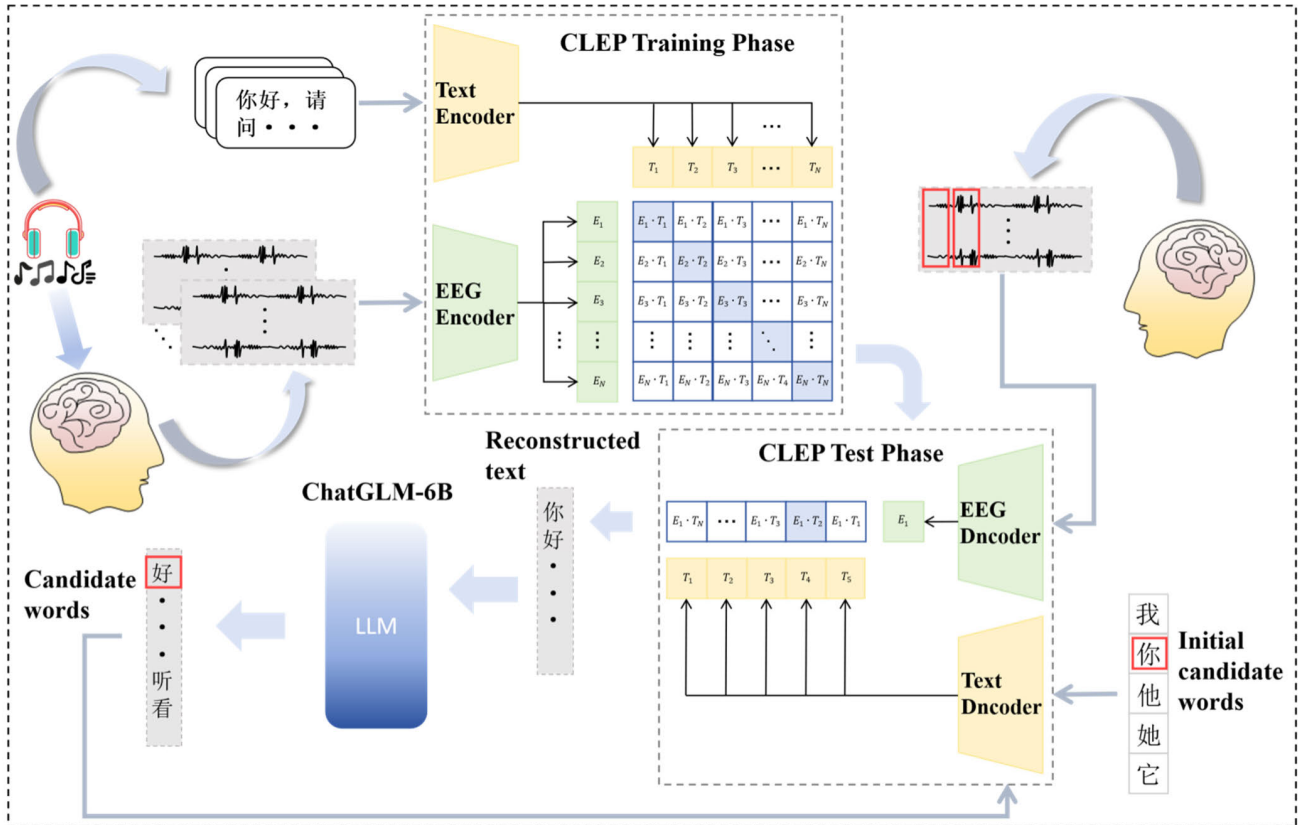
Text Encoder: The text encoder uses Google's open-source model Bert-base-Chinese [29], which is a bi-directional encoder derived from Transformer [30], by which each line of text or each sentence of the textual material is encoded into a 768-dimensional text vector.

EEG encoder: For the EEG encoder, we modified the original CLIP image encoding module. The improved EEG encoding module is shown in Figure 4.

Firstly, this article interprets the maximum length of the EEG data input to the encoder and supplements the parts that are less than the maximum length with two-dimensional zero values (channel dimension and time dimension). After unifying the length of the EEG data, the data is then sent into the EEG encoder. Inside the encoder, without using PSD and WPD feature extraction methods, this article uses two two-dimensional convolution kernels to extract features from the EEG data.

TABLE 1. Structure table of self collected partial EEG data.

Type	Scene	Audio	Audio duration	Record the duration	Number of texts	WLB	JQD	LW	Remark
The original corpus of the last 6-18 Session was generated by artificial intelligence.	Session9	S9-1	10:01	611s	60	(59, 621000)	(59, 626000)	(59, 617000)	Some damaged data has been removed
		S9-2	9:56	606s	60	(59, 613000)	(59, 610000)	(59, 611000)	
		S9-3	10:01	611s	60	(59, 620000)	(59, 615000)	(59, 617000)	
	Session10	S10-1	10:10	620s	60	(59, 626000)	(59, 626000)	(59, 623000)	
		S10-2	9:56	606s	60	(59, 614000)	(59, 616000)	(59, 609000)	
		S10-3	10:02	612s	60	(59, 617000)	(59, 620000)	(59, 621000)	
	Session11	S11-1	10:12	622s	60	(59, 636000)	(59, 627000)	(59, 627000)	
		S11-2	9:22	572s	60	(59, 582000)	(59, 579000)	(59, 577000)	
		S11-3	10:53	663s	60	(59, 671000)	(59, 669000)	(59, 667000)	

**FIGURE 3.** BCI framework for semantic reconstruction based on EEG.

The sizes of the two two-dimensional convolution kernels are (59,1) and (1,80), respectively. After completing feature extraction, a Cls Token is added as a unit of sequence information, followed by position parameters. Then, the entire sequence information is sent into the Transformer Encoder for further feature extraction. Extract through Transformer [30] The unique multi-head self-attention mechanism applies importance levels to each part of the sequence:

$$Attention(Q, K, V) = \text{soft max}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (1)$$

Finally, a 768 * 768 mapping layer is used to achieve dimensional matching between EEG and text.

E. CODING MODEL TRAINING

When training CLEP, the model adopts a contrastive learning based loss function, which is based on the idea of maximizing mutual information. Assuming a query sample x_q , positive samples are x_+ , and the set of negative samples is $\{x_-\}$, the definition of its loss function is as follows:

$$L_{NCE} = -\log \frac{\exp(\text{sim}(x_q, x_+))}{\exp(\text{sim}(x_q, x_+)) + \sum_x \exp(\text{sim}(x_q, x_-))} \quad (2)$$

Here, $\text{sim}(\cdot)$ represents the similarity between sample pairs. By maximizing the similarity of positive samples and minimizing the similarity of sample pairs, the model can learn more discriminative feature representations.

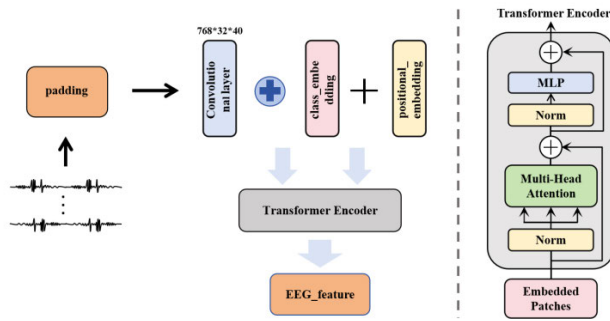


FIGURE 4. EEG encoder.

The training process of the CLEP model is shown in Figure 5.

Using the inner product of text features and EEG features as the cost function, the training objective is to maximize the main diagonal elements of the inner product matrix and minimize other elements. The main diagonal elements are the inner product of paired text and EEG data features, while the remaining elements are the inner product of unpaired text and EEG features. During the experiment, the text input at one end of the text encoder is the text converted from the audio material, and the EEG input at one end of the EEG encoder is the EEG signal recorded under the corresponding listening audio material, the batch size was set to 128. The model was trained on the server using two NVIDIA V100 graphics cards.

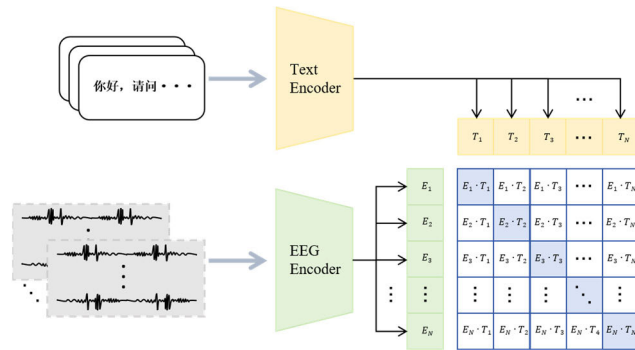


FIGURE 5. Training process of CLEP.

F. DECODING MODEL

The decoding model adopts the Chinese open-source ChatGLM-6B model, which has 6.2 billion parameters, supports bilingual Q&A in both Chinese and English and further optimizes Chinese text generation [31]. The specific decoding process is shown in Figure 6.

Firstly, this article selected five candidate words for the first generated word of the decoded text: “he”, “she”, “it”, “you”, and “I”. By dividing the EEG into segments based on the average time of each character (in this article, the corresponding EEG duration for each character is 0.48s) and inputting them into the CLEP model along with the candidate word text, the model will generate the probability of the best candidate word that matches the EEG data. The candidate

word with the highest probability is selected as the generated word, and the first generated word of the text is obtained. Then, by feeding continuous text into the ChatGLM-6B model, ChatGLM-6B will output the probability of the next candidate word. Here, the study also selects five optimal candidate words and inputs them into CLEP along with the corresponding EEG segment of the next word. Based on their probability of generating candidate words, the subsequent generated text is determined. In the text generation stage, the study adopts a greedy algorithm generation strategy to complete the subsequent text generation until the EEG data is fully input.

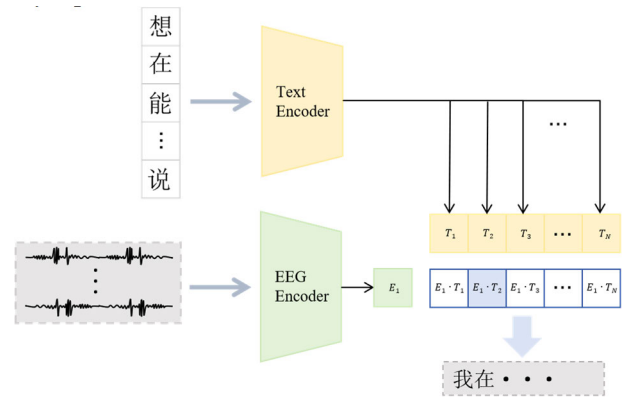


FIGURE 6. Decoding of EEG signals.

IV. RESULT

A. REFACTORING TEXT RESULTS

The paper lists parts of the reconstructed text based on each method under the self-collected data in Table 2, where the labels in the first column represent the corresponding text of the 18th session, and the suffix RR represents the ridge regression method, i.e., the method proposed by Jerry Tang et al [16]. The suffix Random represents the randomized test, and the suffix N-F represents the CLEP model without the feature extraction method.

B. BLEU RATING RESULTS

To quantify the performance, we employed the BLEU metric [32], and the BLEU value is calculated by combining n-gram accuracy and length penalty. The BLEU score formula is:

$$BLEU = BP * \exp\left(\sum_{n=1}^N w_n \cdot \log P_n\right) \quad (3)$$

BP is the length penalty, P_n is the precision of different n-grams (from 1 to N), and w_n is the weight, usually $\frac{1}{N}$, representing the weighting of different n-gram accuracies.

This article conducted BLEU scoring on the reconstructed texts of two types of datasets, and the experimental results are shown in Tables 6.

The results indicate that regardless of the dataset, BLEU-1 and BLEU-2 scores are relatively comparable, while

TABLE 2. Partial reconstructed text based on each method under self-collected data.

Session-18	original text	WLB-RR	WLB-Random	WLB-CLEP-N-F
41	她通过数据分析，不断优化营销策略，提高了广告的投放效果。	它的纸的很面的[SEP][SEP]的	他能做些什么来帮助我？	我喜欢看恐怖故事。请问恐怖故事都有
42	她善于规划，总能合理安排自己的时间和资源，让生活井井有条。	我们的施老子施施[SEP]施施施	你好，请问有什么我可以帮您解答的问题？<0x0A><0x0A>	你是否喜欢这个主题，以及你是否对这个话题感到兴奋和
43	非物质文化遗产承载着民族的记忆，是文化传承的重要组成部分。	你的那我[SEP]很[SEP][SEP][SEP]我	你是否需要休息？-	他喜欢和我一起分享快乐时光和梦想。
44	她利用社交媒体营销，成功提升了品牌的知名度和销量。	我们的我[SEP]我我[SEP]我[SEP]	你是否认为，作为一个公众人物或者一个有	他很喜欢这个过程（我很满意），我很喜欢
45	实施绩效考核制度，激励员工积极进取并提升团队绩效。	你的那万不我的的我[SEP]	他喜欢我，他对我很好，他对我	你是否能够告诉我一些有关地球物理学家这个
46	她热爱学习新知识，不断提升自己的能力和见识。	她们我我[SEP]我[SEP][SEP]	他喜欢照顾人、帮助他人。	他喜欢和我聊天。我应该感到
47	她参与了非物质文化遗产的数字化保护，让古老文化焕发新生。	她的爱我[SEP]我我的[SEP]我	我市开展“美丽家园”建设，旨在	她想要什么，我没有那么容易就可以满足。
48	科学发展促进环保技术创新，为环境保护提供更加高效和可行的解决方案。	它们的包是我的我我我的[SEP][SEP]的我	它（指代上文提到的某个物品或事物）已经具有了很多	你是否喜欢旅行，为什么喜欢/dislike 旅行（仅供参考答案）
49	我每天早晨都会喝一杯温水来唤醒身体。	她的爱很人她的爱	我市开展“美丽家乡”	她（我母亲），一直（从
50	他有着很强的学习能力，总能快速掌握新的知识和技能。	你们的我[SEP]我[SEP]我[SEP][SEP]	她想要什么？她想得到什么样的	他喜欢和我聊天<0x0A>。我感到很高兴

higher-order BLEU scores exhibit poorer performance. We found that regardless of the method used, the results on the self collected dataset (based on the listening audio paradigm) are always better than those on the public dataset (based on the gaze paradigm). Combining our previous analysis of the dataset, we believe that the feature information contained in the EEG signals collected by the listening audio experimental paradigm is significantly higher than that of the gaze paradigm. This finding aligns with the results reported in [17]. So in the subsequent analysis, we will pay more attention to the analysis and discussion of the results of the self collected dataset.

From the comparison of these two indicators(BLEU-1 and BLEU-2), it can be found that on the self-collected dataset, the accuracy of the text recovered by the CLEP model is significantly higher than that of the ridge regression method and random generation.

V. DISCUSSION

A. ADDING FEATURE EXTRACTION METHODS

To improve the accuracy of the recovered text even further, the study attempts to extract frequency domain features from the data by inputting them into the CLEP model before training, and two types of frequency domain feature extraction

methods are selected for the study, which are wavelet packet transform [33] and power spectral density analysis [34]. In the two types of frequency domain feature extraction methods and the model without adopting the frequency domain feature extraction methods, the shape of the original data and the design of the convolution kernel in the model are listed in Table 3.

For PSD, the model does not change the convolution of the first layer, but changes the kernel size of the second layer to (1,4); For WPD, due to the introduction of time-frequency features, the model needs to change the two-dimensional convolution kernel to a three-dimensional convolution kernel and change the size of the second layer convolution kernel to (1, 32, 1).

B. RATING RESULTS OF BLEU COMBINED WITH FEATURE EXTRACTION METHODS

We have recorded the BLEU evaluation scores after incorporating feature extraction methods in Table 4. The scoring results show that the CLEP model with the addition of feature extraction methods does not show significant changes in scores on any dataset. This indicates that the standalone CLEP model has fully captured the implicit features of EEG signals.

TABLE 3. Data shape and convolutional kernel size during feature extraction in two types of dataset.

	Shape of the original data	Feature extraction	After feature extraction	First layer convolution kernel	The second layer convolution kernel
Public data sets Chinese EEG	128*1250	/	128*1250	(128,1)	(1,50)
		WPD	128*32*49	(128,1,1)	(1,32,1)
		PSD	128*41	(128,1)	(1,4)
Self-collected data sets	59*4000	/	59*4000	(59,1)	(1,50)
		WPD	59*32*135	(59,1,1)	(1,32,1)
		PSD	59*41	(59,1)	(1,4)

C. RATING RESULTS OF BERTSCORE COMBINED WITH FEATURE EXTRACTION METHODS

In order to further analyze and demonstrate the accuracy of text recovery, we introduce the BERTscore evaluation method. BERTscore is a language generation evaluation metric based on BERT context embedding, which solves the problems of semantic matching and remote dependency, and has higher relevance to human evaluation. By calculating the cosine similarity of markers, BERTscore can better evaluate the model's ability to capture semantic and sequential information of sentences [35].

The formula of BERTScore:

$$p_{BERT} = \frac{1}{|\hat{x}|} \sum_{\hat{x}_j \in \hat{x}} \max_{x_i \in x} (x_i^T \hat{x}_j) \quad (4)$$

where x is the word embedding for the reference text, and $\hat{x}$ is the word embedding for generating this article.

The final experimental results are shown in Table 5.

In the semantic task of restoring continuous language in the human brain based on EEG, we pay more attention to accuracy. We averaged the accuracy obtained by several subjects under different models. The experimental results showed that on the public dataset of Chinese EEG, the CLEP experiment based on gaze paradigm showed no significant improvement compared to random testing.

The experimental results showed that on the self-collected dataset, the average accuracy obtained by using the CLEP model was 6.2% and 0.9% higher than that obtained by using the ridge regression model and conducting random testing, respectively. We noticed that the average accuracy of the ridge regression model was significantly low. A plausible explanation is that the semantic recovery task for the self-collected EEG data involved longer-duration stimuli compared to the public dataset, and the GPT2 large language model was used in the ridge regression instead of ChatGLM-6B, which has more advantages in Chinese text generation.

In addition, we can fully assume that the main contribution to the improvement in the rating of the CLEP model comes from the CLEP model, rather than ChatGLB-6B, because the

TABLE 4. BLEU scores under different methods on two datasets.

		BLEU Score					
Subject	Method	BLEU-1	BLEU-2	BLEU-3	BLEU-4	BLEU	
Public	Sub04	RR	0.0644	0.0011	0.0000	0.0000	0.0000
		Random	0.0489	0.0007	0.0000	0.0000	0.0000
		CLEP_N_F	0.0426	0.0001	0.0000	0.0000	0.0000
		CLEP_PSD	0.0496	0.0005	0.0000	0.0000	0.0000
		CLEP_WPD	0.0473	0.0003	0.0000	0.0000	0.0000
	Sub08	RR	0.0658	0.0009	0.0000	0.0000	0.0000
		Random	0.0459	0.0001	0.0000	0.0000	0.0000
		CLEP_N_F	0.0466	0.0003	0.0000	0.0000	0.0000
		CLEP_PSD	0.0474	0.0007	0.0000	0.0000	0.0000
		CLEP_WPD	0.0426	0.0000	0.0000	0.0000	0.0000
	Sub13	RR	0.0655	0.0017	0.0000	0.0000	0.0000
		Random	0.0475	0.0005	0.0000	0.0000	0.0000
		CLEP_N_F	0.0439	0.0005	0.0000	0.0000	0.0000
		CLEP_PSD	0.0477	0.0003	0.0000	0.0000	0.0000
		CLEP_WPD	0.0456	0.0004	0.0000	0.0000	0.0000
Self-collected	WLB	RR	0.0753	0.0006	0.0000	0.0000	0.0000
		Random	0.0672	0.0009	0.0000	0.0000	0.0000
		CLEP_N_F	0.0859	0.0003	0.0000	0.0000	0.0000
		CLEP_PSD	0.1000	0.0003	0.0000	0.0000	0.0000
		CLEP_WPD	0.0995	0.0012	0.0000	0.0000	0.0000
	JQD	RR	0.0712	0.0011	0.0000	0.0000	0.0000
		Random	0.0632	0.0006	0.0003	0.0000	0.0000
		CLEP_N_F	0.1029	0.0026	0.0006	0.0003	0.0027
		CLEP_PSD	0.0923	0.0006	0.0000	0.0000	0.0000
		CLEP_WPD	0.0876	0.0009	0.0000	0.0000	0.0000
	LW	RR	0.0679	0.0003	0.0000	0.0000	0.0000
		Random	0.0635	0.0000	0.0000	0.0000	0.0000
		CLEP_N_F	0.0893	0.0009	0.0003	0.0000	0.0000
		CLEP_PSD	0.0807	0.0006	0.0000	0.0000	0.0000
		CLEP_WPD	0.0725	0.0006	0.0000	0.0000	0.0000

random test also uses the ChatGLB-6B generative model, and the rating is significantly lower than that of the CLEP model.

TABLE 5. Z-Test.

	Public dataset Chinese EEG				Self-collected dataset			
	Sub_04	Sub_08	Sub_13	average	WLB	JQD	LW	average
CLEP	0.5486(0.0348)	0.5473 (0.0339)	0.5471 (0.0349)	0.5477	0.5380(0.0315)	0.5470(0.0315)	0.5372(0.0321)	0.5407
WPD_CLEP	0.5475(0.0351)	0.5483(0.0345)	0.5482(0.0340)	0.5480	0.5373(0.0339)	0.5279(0.0369)	0.5210(0.0336)	0.5287
PSD_CLEP	0.5473 (0.0344)	0.5487(0.0340)	0.5490(0.0354)	0.5483	0.5402(0.0301)	0.5363(0.0353)	0.5425(0.0280)	0.5397
Regression	0.5317(0.0376)	0.5339(0.0385)	0.5328(0.0394)	0.5328	0.4591 (0.0273)	0.4586(0.0276)	0.4567(0.0274)	0.4581
Random test	0.5477(0.0281)	0.5477(0.0280)	0.5474(0.0281)	0.5476	0.5332 (0.0286)	0.5333(0.0310)	0.5290 (0.0270)	0.5318

The consistent conclusion can be drawn from the BERTScore score and BLEU score of the CLEP model combined with feature extraction methods: the CLEP model can fully capture the hidden features of EEG signals without combining feature extraction methods.

D. HYPOTHESIS TESTING

We conducted Z-tests on both self collected and dataset data to validate the effectiveness of the CLEP model, with a significance level α set at 0.05. The results are shown in Table 6. The Z-test results show that at a significance level of $\alpha = 0.05$, there is sufficient evidence to suggest that the performance of CLEP exceeds the level of random testing, indicating that the CLEP model is effective.

TABLE 6. Comparison of BERTScore scores after introducing feature extraction methods into CLEP on two datasets.

Z-test ($\alpha=0.05$)		BERTScore
Self-collected dataset	WLB	0.0222
	JQD	$p<0.001$
	LW	$p<0.001$

E. SCALP POTENTIAL DISTRIBUTION PATTERN

This article visualizes the scalp potential distribution of three subjects on a self collected dataset. Specific method: On the data collected by three subjects, 1. average the EEG data of each test and time based on the paradigm of listening materials, 2. average the EEG data of three subjects in the resting state (about eight seconds), and finally draw the brain map, as shown in Figure 7. The three subgraphs in the first row of Figure 7 reflect the brain activity when listening to audio. The brain reactions of these three subjects are concentrated in the left brain region, while the three subgraphs in the second row reflect the brain activity in the resting state. It can be seen that the brain activity in the task state (listening to audio), especially in the left brain region, shows a more obvious difference, which is reflected in the degree of activity reflected by color. The left brain area includes the functional areas of human brain processing and language production, namely Broca area and Wernick area.

Specifically, Broca's area is located in the frontal lobe of the brain, in areas 44 and 45 behind the inferior frontal

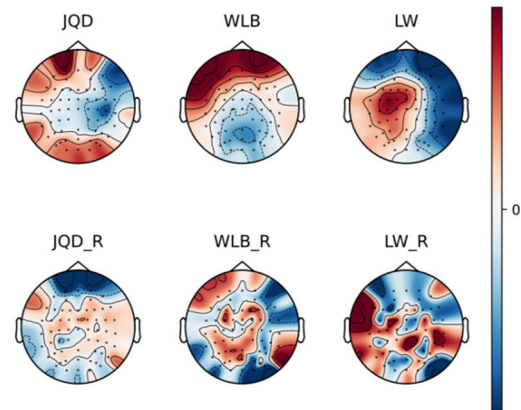


FIGURE 7. Brain mapping of EEG signals.

gyrus of the cerebral cortex, near the lateral fissure of the brain. This area, along with Wernicke's area located in the left temporal lobe, specifically in the posterior superior temporal gyrus (Brodmann 22), near the auditory cortex, forms the language center and is an important part of the human brain responsible for processing and generating language information. This is consistent with the expected brain responses generated by applying audio stimuli to the subjects in the experiment.

F. EEG TRACEABILITY

Finally, this article traces the origin of EEG. The experiment utilized sLORETA technology to analyze the significant differences in resting state and task state (listening to audio) EEG data of three subjects in the self collected data. The visualization results are shown in Figure 8.

Observing Figure 8, it can be observed that significant differences in brain regions were observed between the resting and task states in all three subjects, confirming the differences in EEG signals generated by the subjects in the two scenarios. In the tracing diagram of the WLB of the subjects, it can be found that the brain regions with significant differences come from the auditory cortex and Wernicke area, which contain the language receiving and processing functional areas of the human brain. Specifically, the auditory cortex is mainly located deep in the lateral fissures of the temporal lobes on both sides of the brain. It is the highest level center of our auditory perception, responsible for converting neural signals from the ears into the sound world that we can perceive

and understand. The Wernick area is located in the posterior temporal lobe of the left hemisphere of the brain and is a key center for language comprehension. The auditory cortex and Wernicke's area together form the language center, which is an important part of the human brain responsible for receiving and processing language information.

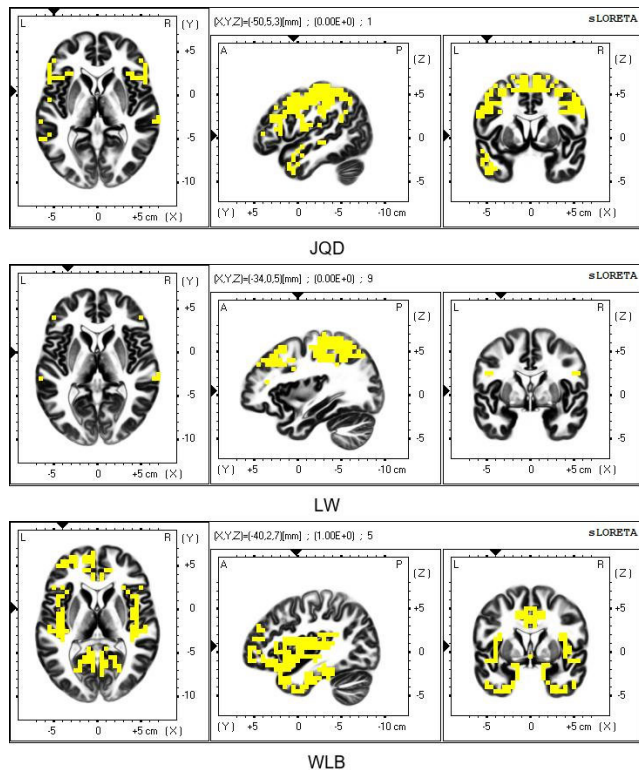


FIGURE 8. Brain region traceability difference map based on self-collected dataset.

Due to individual differences in EEG signals, no significant differences in language functional areas were observed in the source tracing maps of the other two subjects. However, it was confirmed that there were significant differences in the EEG signals presented by the subjects in the two states, and this also confirmed to some extent the characteristic of whole brain cooperation in processing audio signals.

VI. CONCLUSION

This paper proposes an EEG-based BCI framework for reconstructing continuous semantic content from brain activity, incorporating a novel CLEP model. The framework recovers text by preprocessing the EEG signal, by CLEP's classification discrimination, and then combines it with the candidate word generation of the large language model ChatGLM-6B. In order to fully prove the effectiveness of the framework and the model, in addition to validating it on the publicly available dataset Chinese EEG, the study also builds an EEG dataset under the listening to audio materials.

The rating shows that the data evaluation results based on the listening audio experimental paradigm are better than those based on the fixation paradigm. CLEP has been proven to be effective at a significance level of $\alpha = 0.05$. In addition, experiments with the added feature extraction method have also demonstrated that the proposed CLEP model has fully captured the time-frequency features of the original EEG signals.

Currently, the method has only been trained and tested on offline data. For real-time decoding, further improvement in accuracy is needed before it can be considered. In addition, although EEG data has been preprocessed by filtering and downsampling, further research is required to mitigate the noise interference inherent in online (real-time) data acquisition. Finally, if this method wants to generalize to different subjects, it is also a huge challenge, because the individual differences in EEG signals have always been an unresolved difficulty in the computer field.

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